

A counterexample to strong peak density in $A_u(B_X)$ under analytic RNP

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Source Question

Choi, Lee, and Song [1, p. 16] ask whether, if a complex Banach space X has the analytic Radon–Nikodým property, the set of all strong peak functions in either $A(B_X)$ or $A_w(B_X)$ is dense. In their notation $A(B_X)$ denotes either $A_b(B_X)$ or $A_u(B_X)$.

$A_w(B_X) = \{f \in A_b(B_X) : f \text{ is weakly continuous on } B_X\}.$

We shall denote by $A_w(B_X)$ one of $A_{wb}(B_X)$ and $A_{wu}(B_X)$. The proof of Theorem 4.4 and Remark 4.5 show that the set of all strong peak functions for $A_w(B_X)$ is dense in $A_w(B_X)$ if X has the Radon–Nikodým property.

It is a natural question that the set of all strong peak functions in either $A(B_X)$ or $A_w(B_X)$ is dense, if X has the *analytic Radon–Nikodým property*. The answer is negative in $A_w(B_X)$ as observed in [30]. Recall that a complex Banach space X is said to have the *analytic Radon–Nikodým property* if for every bounded analytic function f from the open unit disc of $\mathbb{C}$ into X , it has the a.e. radial limits

$$f(e^{i\theta}) = \lim_{r \uparrow 1} f(re^{i\theta}) \quad \text{a.e. } \theta.$$

For more details on the analytic Radon–Nikodým property, see [11, 30].

Notice that $L_1[0, 1]$ is uniformly c -convex and has the analytic Radon–Nikodým property (cf. [38]). Let $X = L_1[0, 1]$. We shall show that $A_w(B_X)$ does not contain any strong peak function. Indeed, suppose that $f \in A_w(B_X)$ is a strong peak function at x . For each $n \geq 1$, let

The same passage records that the weak variant $A_w(B_X)$ is negative for $X = L_1[0, 1]$. The result below gives a negative answer for the uniformly continuous analytic algebra $A_u(B_X)$.

Idea of the Proof

The obstruction is not weak continuity built into the algebra, but weak sequential continuity forced by uniform continuity plus the Dunford–Pettis property of L_1 . Scalar polynomials on L_1 are weakly sequentially continuous, and polynomials are uniformly dense in $A_u(B_X)$. Thus every A_u -function is weakly sequentially continuous on the ball.

On the other hand, the Rademacher functions split any $x \in S_{L_1}$ into rapidly oscillating halves. The normalized sequence $(1 + r_n)x / \|(1 + r_n)x\|_1$ converges weakly back to x , but remains about distance 1 from x . A weakly sequentially continuous function therefore cannot strongly peak at x .

Main Result

Theorem. *Let $X = L_1[0, 1]$, complex-valued. Then $A_u(B_X)$ contains no strong peak function. Consequently, although X has the analytic Radon–Nikodým property, the strong peak functions are not dense in $A_u(B_X)$.*

Proof. We first record that every $f \in A_u(B_X)$ is weakly sequentially continuous on B_X . Indeed, $L_1[0, 1]$ has the Dunford–Pettis property. A standard consequence is that every scalar continuous polynomial on $L_1[0, 1]$ is weakly sequentially continuous on bounded sets; equivalently, scalar polynomials are completely continuous on weakly convergent sequences [2]. Also, continuous polynomials are uniformly dense in $A_u(B_X)$: for $f \in A_u(B_X)$, the functions $x \mapsto f(rx)$, $0 < r < 1$, converge uniformly to f as $r \uparrow 1$, and each $x \mapsto f(rx)$ is uniformly approximated on B_X by its Taylor polynomials. Hence a uniform approximation argument gives weak sequential continuity of f on B_X .

Suppose, toward a contradiction, that $f \in A_u(B_X)$ is a strong peak function at $x \in B_X$. A nonconstant holomorphic function on the open unit ball cannot attain a strong maximum at an interior point: restricting to any complex line through an interior point and using the maximum modulus theorem would force local, hence global, constancy on the connected ball. Thus $\|x\|_1 = 1$.

Let (r_n) be the Rademacher functions on $[0, 1]$. Since $r_n \rightarrow 0$ weak-star in $L_\infty[0, 1]$, for every $h \in L_1[0, 1]$ we have $\int r_n h \rightarrow 0$. Put

$$a_n = \|(1 + r_n)x\|_1 = \int_0^1 (1 + r_n)|x| dt.$$

Then $a_n \rightarrow 1$, so $a_n > 0$ eventually. Passing to this tail, define

$$y_n = \frac{(1 + r_n)x}{a_n} \in S_X.$$

For every $g \in L_\infty[0, 1]$,

$$\int gy_n = \frac{1}{a_n} \int g(1 + r_n)x \longrightarrow \int gx,$$

so $y_n \rightarrow x$ weakly in $L_1[0, 1]$. By the weak sequential continuity just proved, $f(y_n) \rightarrow f(x)$. Since f peaks at x ,

$$|f(y_n)| \longrightarrow |f(x)| = \|f\|.$$

Strong peaking would imply $y_n \rightarrow x$ in norm.

But this is false. We have

$$\begin{aligned} \|y_n - x\|_1 &\geq \|(1 + r_n)x - x\|_1 - \|y_n - (1 + r_n)x\|_1 \\ &= \|r_n x\|_1 - |1 - a_n| \longrightarrow 1. \end{aligned}$$

This contradicts the defining sequential condition for a strong peak function. Therefore $A_u(B_{L_1[0,1]})$ has no strong peak functions.

Finally, $L_1[0, 1]$ has the analytic Radon–Nikodým property, as recalled in the source passage [1, p. 16]. Hence this is a counterexample to density of strong peak functions in $A_u(B_X)$ under analytic RNP. $\square$

Limitations and Novelty Check

This result is deliberately scoped to $A_u(B_X)$. It does not prove that the larger algebra $A_b(B_X)$ has no strong peak functions. The proof uses uniform continuity through polynomial approximation, and this step is not available verbatim for all of $A_b(B_X)$.

Before promotion, the run indexes were searched for 0708.4069, Bishop’s theorem, analytic Radon–Nikodým, strong peak functions, $A_u(B_X)$, and $L_1[0, 1]$. Web searches for exact phrases such as “no strong peak function $A_u(B_{L_1})$ ” and “strong peak analytic Radon–Nikodým $A_u(B_X)$ ” found the source paper and nearby weakly-uniform/numerical negative results, but no exact prior run packet for this scalar A_u counterexample.

Human review should focus on the quoted standard polynomial fact and the uniform-density step for $A_u(B_X)$; these are classical, but they are the only external functional-analytic inputs in the argument.

References

- [1] Y. S. Choi, H. J. Lee, and H. G. Song, *Bishop’s Theorem and Differentiability of a subspace of $C_b(K)$* , arXiv:0708.4069.
- [2] J. Farmer and W. B. Johnson, *Polynomial Schur and polynomial Dunford–Pettis properties*, Contemporary Mathematics **144** (1993), 95–105.